

Back analysis of rock mass parameters in tunnel engineering using machine learning techniques

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ABSTRACT

Efficient determination of the rock mass properties is vitally important for calculating and evaluating tunnel stability in tunnel engineering. The back analysis method has been widely used as an indirect method for determining rock mass parameters based on field measurements. However, most back analysis methods are generally time-consuming for numerical simulation and are merely based on the measured displacement, which leads to the identification of rock mass parameters that cannot fully reflect the characteristics of the surrounding rock. To improve the accuracy of the estimation of rock mass parameters, this paper presents a back analysis method based on multi-output support vector regression (MSVR) and differential evolution (DE) algorithms. Firstly, the global sensitivity analysis of rock mass parameters is analyzed using the elementary effects method. Numerical simulation is then carried out to prepare training samples. DE algorithm is used to determine the optimum hyperparameters of MSVR. Based on the monitoring data, the rock mass properties of the selected sensitive parameters are estimated by the constructed MSVR model. A high-speed railway tunnel is utilized to demonstrate the effectiveness of the MSVR with the DE algorithm (DE-MSVR). The results show that the DE-MSVR with mixed monitoring data of vault settlement, convergence, and rock mass stress has higher forecasting performance than these models with a single type of monitoring data. It is feasible to use the monitoring data at the early stages combined with the numerical simulation for parameter back analysis. Moreover, the comparison results show that the presented method exhibits higher prediction accuracy than the existing back analysis models.

1. Introduction

The high-speed railway (HSR) is emerging to play the main role in mass transport systems because of its safety, speed, bearing capacity, and sustainability (Zhang et al., 2016; Zhang et al., 2019). As a vital underground part, the HSR tunnel has been increasing dramatically, especially for tunnel projects using the New Austrian Tunneling Method (NATM) (Guan et al., 2013). The rock mass properties predominantly influence the stability of soil and rock environments, especially in complicated underground constructions (e.g., fault, high groundwater level, soft soil, and low overburden). To understand the structural behavior and numerically model the real conditions, various input parameters related to the rock mass are required, such as Young's modulus and cohesion. However, because of the high heterogeneity, nonlinearity,

and discontinuity of the rock mass, the determination of mechanical parameters is always a perennial challenge. The use of geotechnical parameters with low accuracy in the analysis of structural stability could cause unexpected safety problems during construction, such as cavern collapses, vault cracking, and even geological disasters (Chang et al., 2022; Sharifzadeh et al., 2013). Therefore, accurate estimation of rock mass properties is essential to simulate the actual geological conditions, understand the structural behavior, and make timely adjustments to the construction design (An and Song, 2018; Gao and Ge, 2016; Wu et al., 2019).

Nowadays, the back-analysis method has been proven to be one of the most useful methods to estimate the geomechanical parameters of rock and soil environments for the safe execution of geotechnical structures (An et al., 2020; Li et al., 2020a; Mahmoudi and Rajabi, 2020). The back-analysis method can use monitoring data during

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Nomenclature			
$\mathbf{x}_i$	Input vector of MSVR	$\mathbf{X}_s^{//g}$	Trial vector at the gth generation
$\mathbf{y}_i$	Output vector of MSVR	ρ	Uniformly distributed random number within [0, 1]
$\mathbf{W}, \mathbf{b}$	Regressor vectors	P_{cr}	Probability factor that controls the population diversity
$\mathbf{e}_i^T$	Error equation	$\mathbf{x}_i^v$	Model input of EE with d input factors ($v = 1 \cdots d$)
γ	Parameter that represents the trade-off between the regularization and the error reduction term	u	Number of selected levels
CT	Constant term	σ	Sample mean
u_i, a_i	Iteration parameters	μ	Sample standard deviation
$\beta, \mathbf{b}$	Best parameters of MSVR in feature space	μ^*	Revised version of the measure μ
$\mathbf{K}$	Kernel matrix	$\hat{\mathbf{y}}_i$	Estimated values
r	Kernel parameter of RBF kernel function	l	Number of samples
$\mathbf{X}_s$	Initial population of DE	MSVR	Multi-output support vector regression
N_P	Number of the initial population	DE	Differential evolution algorithm
D	Decision variables	LS-SVR	Least-squares support vector regression
$x_p^{\min}, x_p^{\max}$	Lower and upper bounds of the p th decision variable	GPR	Gaussian process regression
$\mathbf{X}_s^g$	Target vector at the gth generation	BPNN	Backpropagation neural network
$\mathbf{X}_s^{//g}$	Mutant vector at the gth generation	RMSE	Root-mean-squared error
		MAPE	Mean absolute percentage error

Table 1
Summary of back-analysis methods.

Back-analysis method	Pros	Cons
Analytical solution	Direct mathematical formula, clarity, comprehensibility, rapid response, generalizability	Rough approximation, simplifying assumptions, limited applicability, boundary condition constraints
Numerical simulation	Setting complex boundary conditions, trial and error method, cost-effective, flexibility, iteration	Tedious modeling, time-consuming computation, uncertainties, sensitivity to initial conditions
ML techniques	High-order approximation, pattern recognition, multiple algorithms, efficient computation, better accuracy, automation	Data dependency, large number of training samples, local optimum

structural construction in addition to the results of in-situ tests and laboratory tests (Ghorbani et al., 2020). It generally consists of a forward procedure and a backward procedure (Wu et al., 2019). The forward procedure establishes a mapping between rock mass parameters and the monitoring data. The backward procedure then adjusts and estimates the rock mass parameters by minimizing the error between the computed data and the measured data (Li et al., 2006).

Three types of back-analysis methods have been established for underground structures, including analytical solutions, numerical simulations, and machine learning (ML) techniques (Zhang et al., 2020; Zhu et al., 2014). Table 1 shows the summary of back-analysis methods. The analytical solution relates rock mass properties to the tunnel deformation directly based on the traditional convergence-confinement method, and it is a rough approximation to geotechnical engineering (Chu et al., 2019; Wang et al., 2018; Zhu et al., 2014). This method can hardly be used in real tunnel projects when the boundary conditions do not fully adhere to their assumptions (Deng et al., 2017). The second category is numerical simulation, such as the finite element method, discrete element method, and boundary element method. Numerical simulation can set complicated boundary conditions and theoretically obtain any design parameters by trial and error (Khamesi et al., 2015; Sharifzadeh et al., 2013; Zhang et al., 2020). In fact, the numerical model is generally time-consuming as it needs to be simulated many times in the

optimization procedure, especially when the convergence of the back procedure is slow (Zhuang et al., 2019). ML techniques can provide high-order approximations for nonlinear problems (Li et al., 2021; Torabi-Kaveh and Sarshari, 2019). To improve the computational efficiency of numerical simulation, ML techniques have been used in the back analysis method to establish the relationship between rock mass parameters and structural responses. The current studies mainly use the measured displacement at the stable stages to estimate the rock mass parameters, which means that it demands several months to obtain the monitoring data (Gao and Ge, 2016; Hajihassani et al., 2019; Sun et al., 2021). Thus, when the monitoring data are ready, the estimation results of the rock mass parameters may be useless because of the rapid tunnel construction. Moreover, although satisfactory results were obtained, neural network algorithms generally need a large number of training samples to obtain satisfactory performance (Fei et al., 2020). It is difficult to train a high-capacity model without overfitting when performed on small sample problems (Ding et al., 2020).

In the ML community, support vector regression (SVR) has been considered a highly promising approach for nonlinear predictive tasks, especially for small sample problems (Skentou et al., 2022; Zhu et al., 2014). Compared with neural network algorithms, the computational complexity of SVR is not dependent on the dimensionality of the input space, and multiple local extreme values can be avoided (Cantillo-Luna et al., 2022; Zhang and Wang, 2023; Zhu and Gao, 2018). Despite these advantages, the traditional SVR only contains a one-dimensional output, which mainly addresses single-output problems. For multi-output regression problems, SVR needs to transform these problems into a set of independent single-output regression problems and solve them independently (Chen et al., 2021). However, it immensely increases the total computation time. It also disregards the potential correlations between the multiple outputs and possibly results in inaccurate predictions (Li et al., 2020b). Multi-output support vector regression (MSVR), proposed by Tuia et al. (2011), is a generalization of SVR that can handle multi-output regression problems accurately. MSVR has been used in a variety of disciplines, including communication (Sanchez-Fernandez et al., 2004), energy market (Cantillo-Luna et al., 2022), and material modeling (Zhao et al., 2015). MSVR retains superior generalization capacities in dealing with small sample problems and could handle multi-output regression problems efficiently (Tuia et al., 2011). Thus, it could be more suitable for estimating the rock mass properties of tunnel projects.

The selection of hyperparameters directly influences the forecasting

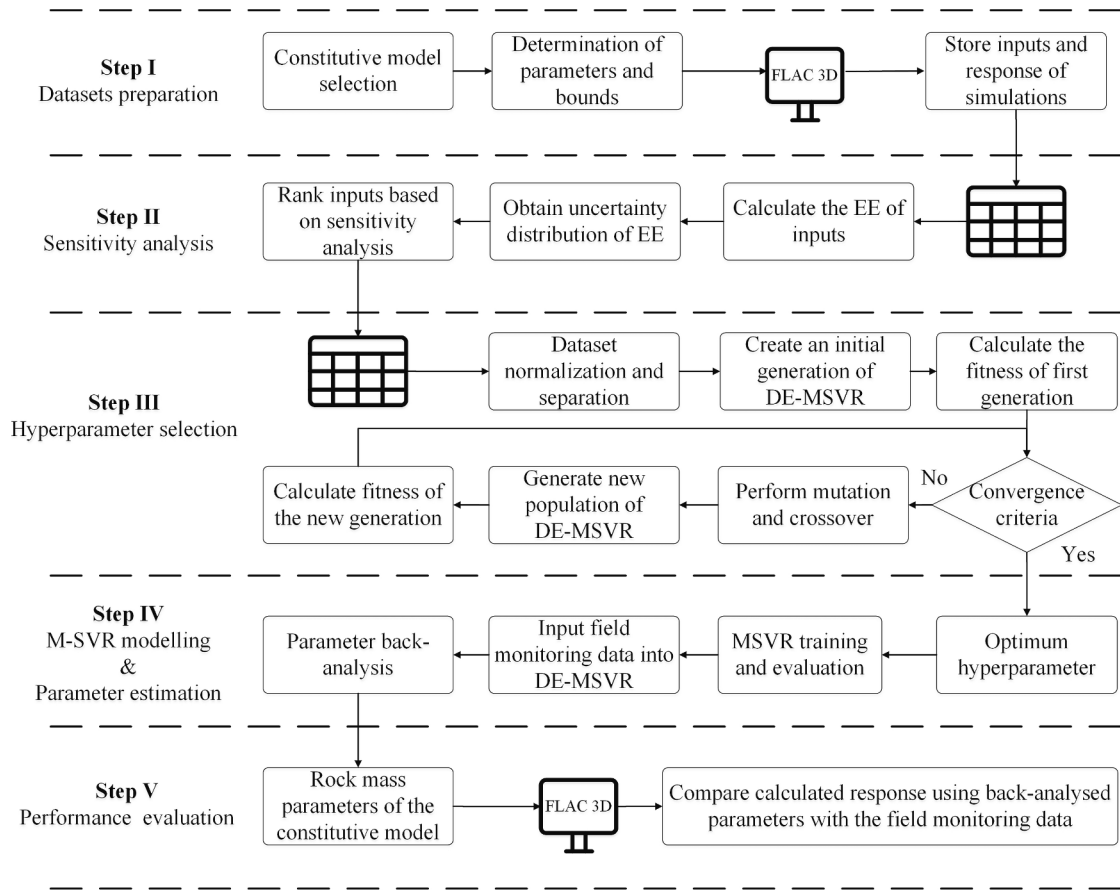


Fig. 1. The procedure of the back analysis method.

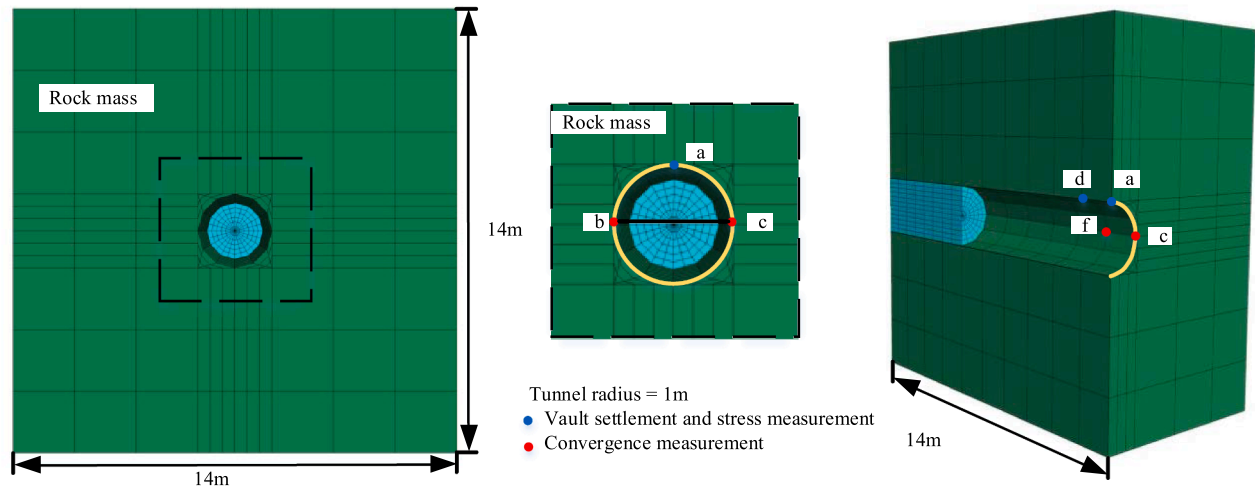


Fig. 2. The finite difference model and measuring points.

Table 2

Lower/upper bounds of the rock mass parameters.

Density ($\text{kg}\cdot\text{m}^{-3}$)	Young's modulus (MPa)	Poisson's ratio	Friction angle ($^{\circ}$)	Cohesion (kPa)
1900–2100	50–500	0.2–0.3	25–35	80–100

performance of MSVR. The task of selecting the optimal hyperparameters from a set of candidates is called model selection (Le et al., 2022; Luo et al., 2018). Many nature-inspired algorithms have been introduced to search for the optimal hyperparameters of ML algorithms, such as genetic algorithm, particle swarm optimization, simulated annealing, imperialism competitive algorithm, whale optimization algorithm, and artificial fish swarm algorithm (Gao, 2020; Harandizadeh et al., 2021; Qi and Fourie, 2018). These algorithms also solved the back analysis problems in previous studies (Ghorbani et al., 2020; Vardakos

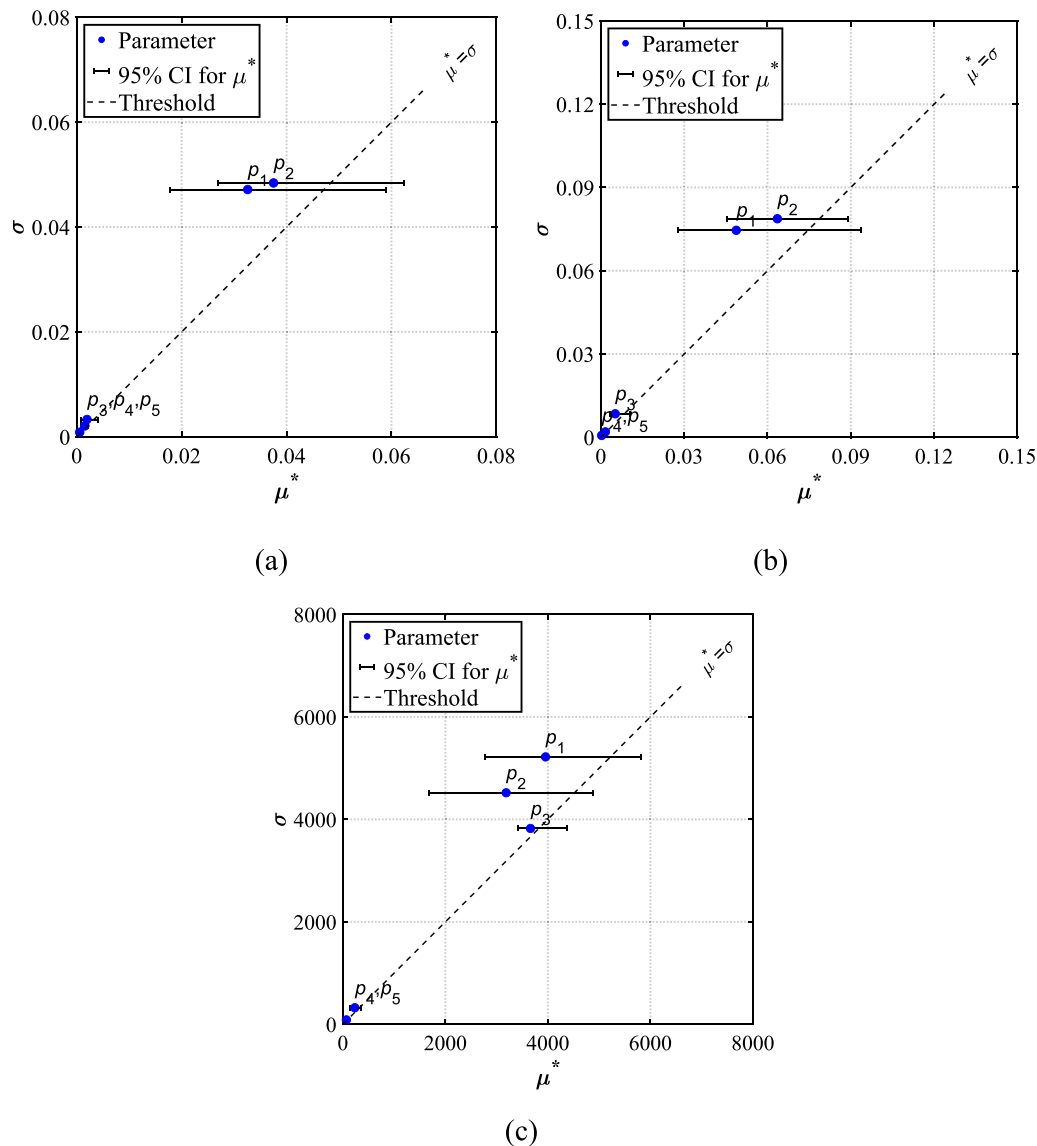


Fig. 3. Sensitivity analysis results for No.1 monitoring section (a) vault settlement (b) convergence and (c) surrounding rock stress. (p_1 : density, p_2 : Young's modulus, p_3 : Poisson's ratio, p_4 : cohesion, and p_5 : frictional angle).

et al., 2016; Zhang et al., 2020). However, in most studies, only one optimization algorithm was used for their specific problems, and optimization results may be hugely different for different optimization algorithms (Gao, 2020). Thus, it is necessary to compare the accuracy of nature-inspired algorithms to select a suitable algorithm and apply the algorithms appropriately.

In summary, the main contributions of this paper are two-fold. First, a new back analysis method based on MSVR and differential evolution (DE) algorithm is proposed to estimate rock mass properties. The present method exhibits superior generalization capacities in handling small sample problems and significantly improves the prediction accuracy of the estimation of rock mass properties. To the best of the authors' knowledge, the MSVR has rarely been used in underground structures, and this is the first time to apply it for estimating rock mass properties. Second, the feasibility of the proposed method is validated using monitoring data collected from a high-speed railway tunnel. To obtain rock mass properties efficiently and accurately, different types of monitoring data are used to estimate the parameters of the rock mass, including vault settlement data, convergence data, stress data, and these data for different tunnel face advancements. Results show that the DE-MSVR with mixed monitoring data of vault settlement, convergence,

and rock mass stress has higher performance than these models with a single type of monitoring data. It is feasible to use the monitoring data at the early stages combined with the numerical simulation for parameter back analysis. Moreover, the presented model exhibits higher accuracy than the existing back analysis models.

The rest of this article is organized as follows. In Section 2, a new back analysis method based on MSVR and DE algorithm is presented. Section 3 validates the performance of the presented model. In Section 4, the effectiveness of the present approach is demonstrated by the monitoring data collected from a high-speed railway tunnel under construction. A comparison of four models is also performed. Finally, the concluding remarks of this study are presented in Section 5.

2. Back Analysis Method Based on MSVR and DE

2.1. Brief introduction to MSVR

MSVR is a generalization of the standard SVR with the basic idea of extending the ϵ -insensitive loss function to multi-dimensional outputs and handling all outputs simultaneously (Bao et al., 2014). Detailed discussions on the MSVR can be found in Tuia et al. (2011), and only a

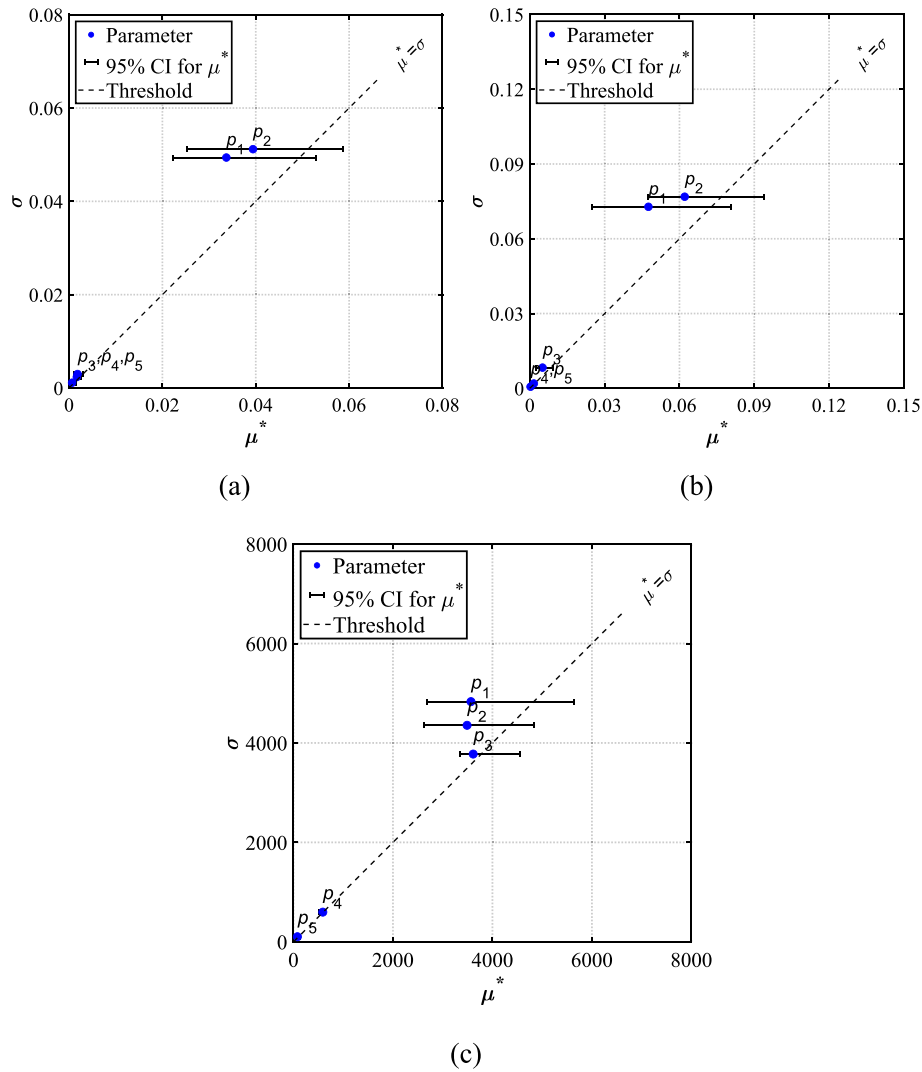


Fig. 4. Sensitivity analysis results for No.2 monitoring section (a) vault settlement (b) convergence and (c) surrounding rock stress. (p_1 : density, p_2 : Young's modulus, p_3 : Poisson's ratio, p_4 : cohesion, and p_5 : frictional angle).

Table 3
Selected rock mass parameters and back-analysis target parameters.

Parameter	Lower	Upper	Target
Density ($\text{kg}\cdot\text{m}^{-3}$)	1900	2100	2024
Young's modulus (MPa)	50	500	175
Poisson's ratio	0.2	0.3	0.24

Table 4
Training and testing results of the back analysis model for five nature-inspired algorithms.

Method	GA-MSVR	POS-MSVR	DE-MSVR	SA-MSVR	SAFSA-MSVR
Training error	0.00480	0.00473	0.00343	0.00494	0.01350
Testing error	0.00833	0.00847	0.00566	0.00829	0.01997

brief introduction to MSVR is provided here.

Given a set of training samples $\{x_i, y_i\}_{i=1}^l$ where $x_i = [x_1, x_2, \dots, x_d]$ is the input vector and $y_i = [y_1, y_2, \dots, y_Q]$ is the output value, MSVR is formulated as a minimization of the following function (Mao et al.,

Table 5
Target and estimation values of rock mass parameters for five nature-inspired algorithms.

Parameter	Density ($\text{kg}\cdot\text{m}^{-3}$)	Young's modulus (MPa)	Poisson's ratio	RMSE	MAPE (%)
Target	2024.00	175.00	0.24000	–	–
GA-MSVR	2025.24	176.95	0.24019	0.00450	0.41858
POS-MSVR	2025.26	177.05	0.24023	0.00468	0.44241
DE-MSVR	2025.20	176.51	0.24028	0.00430	0.34758
SA-MSVR	2025.26	177.04	0.24022	0.00466	0.43954
SAFSA-MSVR	2024.82	171.44	0.23894	0.00799	0.83855

Table 6
Comparison results of prediction monitoring data for five algorithms.

Method	GA-MSVR	POS-MSVR	DE-MSVR	SA-MSVR	SAFSA-MSVR
RMSE	0.2011	0.1962	0.1369	0.1919	0.3685
MAPE	0.9377	0.9162	0.6455	0.8911	1.7126
A5	95.70%	96.60%	97.65 %	96.02%	93.98%

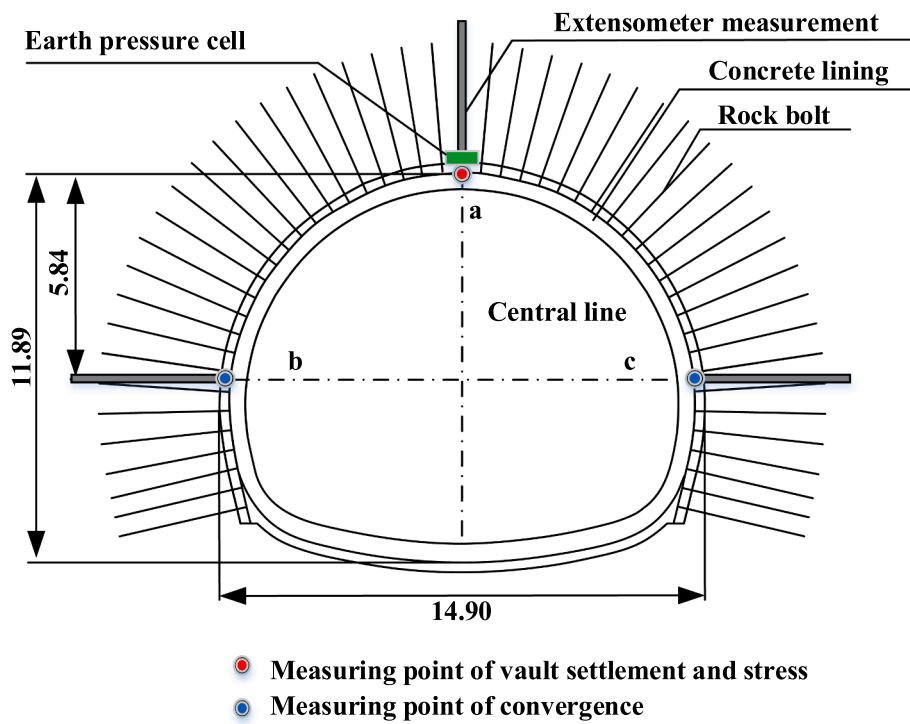


Fig. 5. A typical cross-section and three measuring points for a tunnel section of the Yangshan tunnel (Unit: m).

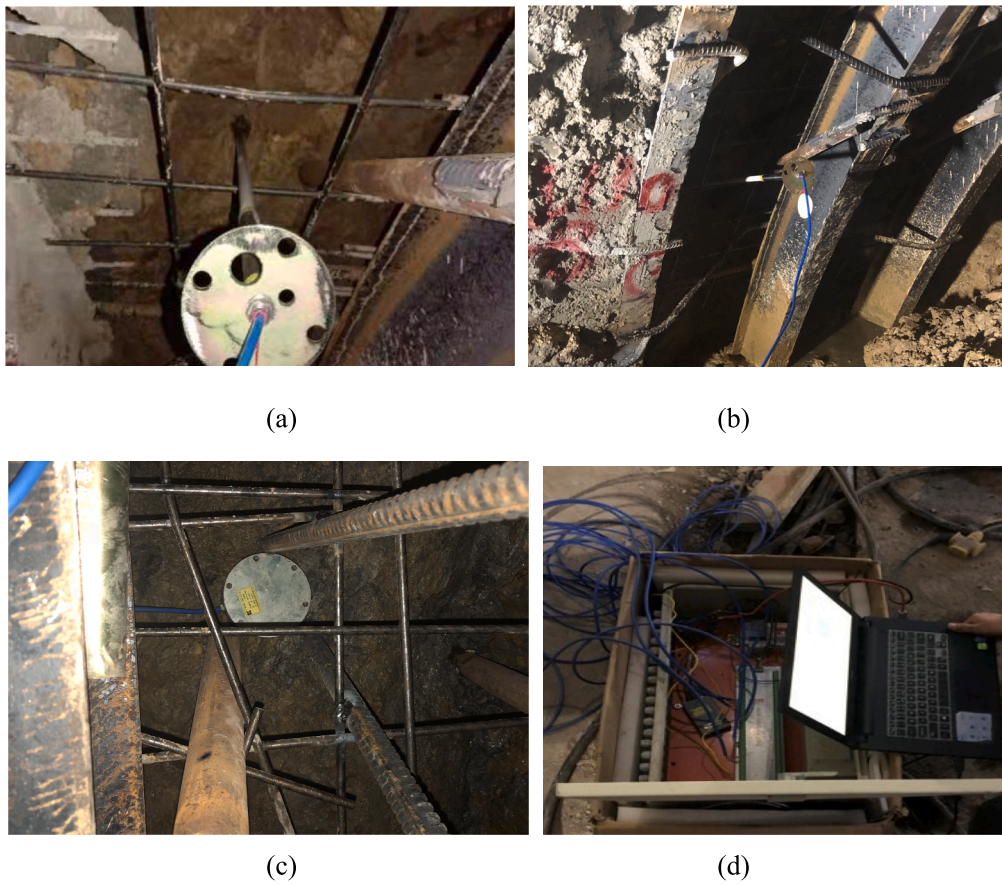


Fig. 6. Monitoring instrument installation during the construction of the Yangshan tunnel (a) multipoint extensometer measurements (b) multipoint extensometer measurements (c) earth pressure cell and (d) data collection.

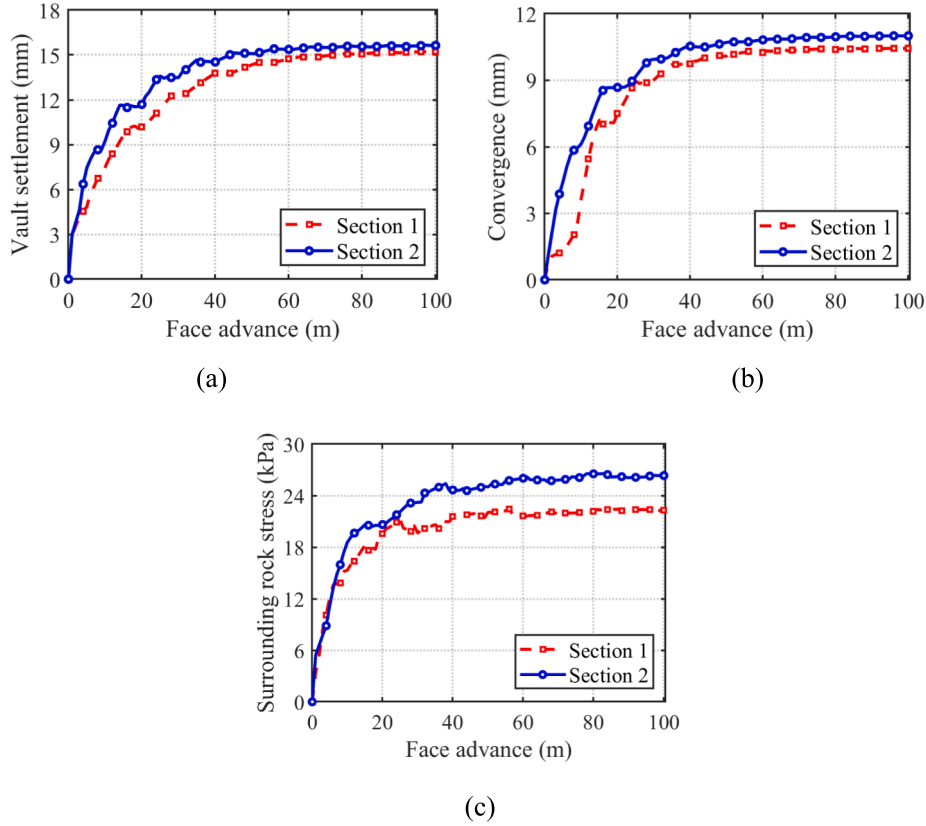


Fig. 7. Monitoring data of two tunnel monitoring sections for the Yangshan tunnel (a) vault settlement (b) convergence and (c) surrounding rock stress.

Table 7

The mechanical parameters of the tunnel support.

Material type	Young's modulus (MPa)	Poisson's ratio
Concrete lining	29,000	0.2
Rock bolt	206,000	0.3

2014):

$$\min_{\mathbf{W}, \mathbf{b}} Lp = \frac{1}{2} \sum_{j=1}^Q \|\mathbf{w}^j\|^2 + \gamma \sum_{i=1}^l L(u_i) \quad (1)$$

where $\mathbf{W} = [\mathbf{w}^1, \dots, \mathbf{w}^Q]$; $\mathbf{b} = [b^1, \dots, b^Q]^T$; $\mathbf{e}_i^T = \mathbf{y}_i^T - \phi^T(\mathbf{x}_i)\mathbf{W} - \mathbf{b}^T$; $\phi(\cdot)$ is a mapping function to the feature space which is a higher dimensional space; γ is a parameter that represents the trade-off between the regularization and the error reduction term; and $L(u_i)$ is a differentiable version of the quadratic ε -insensitive cost function (Tuia et al., 2011):

$$L(u_i) = \begin{cases} 0, & u_i < \varepsilon \\ u_i^2 - 2u_i\varepsilon + \varepsilon^2, & u_i \geq \varepsilon \end{cases} \quad (2)$$

where $u_i = \|\mathbf{e}_i\| = \sqrt{\mathbf{e}_i^T \mathbf{e}_i}$.

By adopting the cost function $L(u_i)$ described in Eq. (2), when ε is nonzero, MSVR considers all outputs to construct each individual regressor and gets a robust solution based on the information. It should be noted that Eq. (1) cannot be solved straightforwardly. Then, an iterative reweighted least squares (IRWLS) procedure is used to obtain the desired solution (Sanchez-Fernandez et al., 2004). By introducing a first-order Taylor expansion of the cost function $L(u_i)$, the objective of Eq. (1) will be approximated and can be expressed as follows (Li et al., 2020b):

$$Lp'(\mathbf{W}, \mathbf{b}) = \frac{1}{2} \sum_{j=1}^Q \|\mathbf{w}^j\|^2 + \frac{1}{2} \sum_{i=1}^l a_i u_i^2 + CT \quad (3)$$

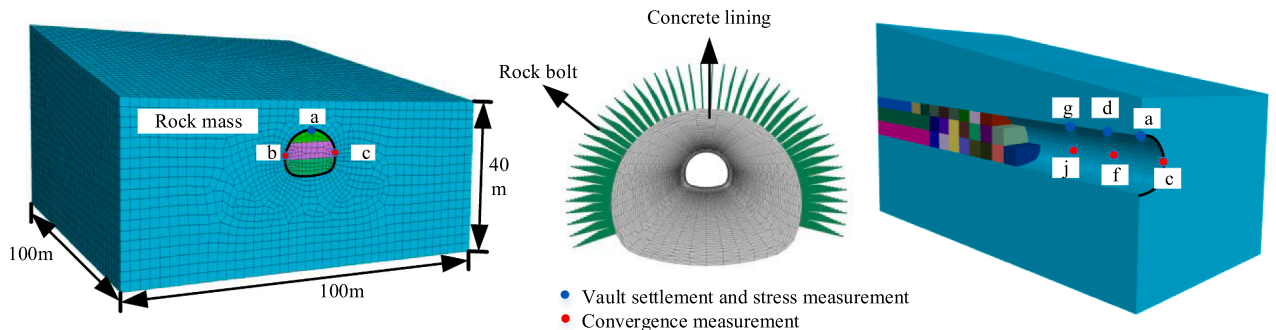


Fig. 8. The finite difference model of the Yangshan tunnel and measuring points.

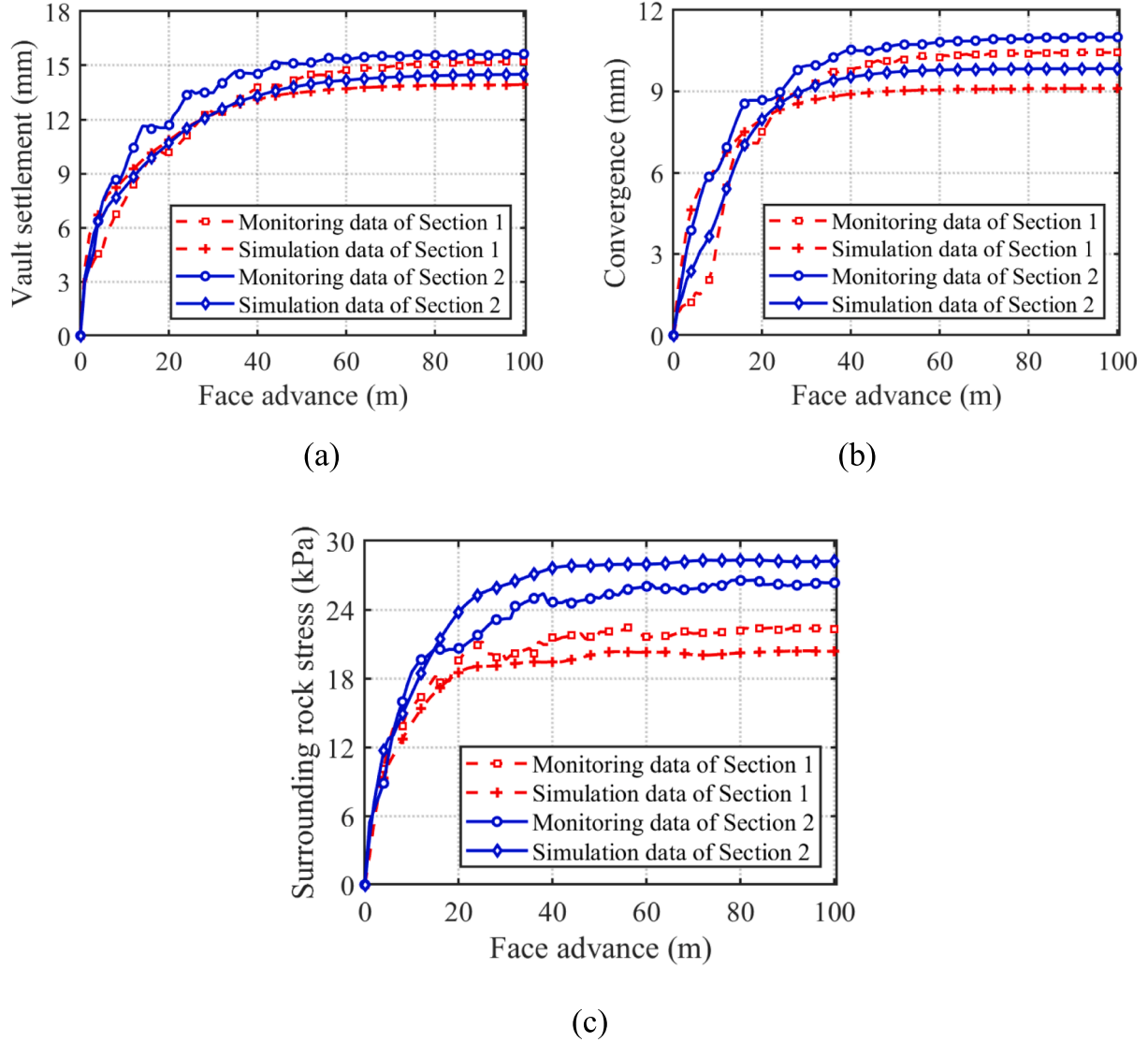


Fig. 9. Comparison between the numerical solution and the measuring data (a) vault settlement (b) convergence and (c) surrounding rock stress.

$$a_i = \begin{cases} 0, & u_i^k < \varepsilon \\ \frac{2\gamma(u_i^k - \varepsilon)}{u_i^k}, & u_i^k \geq \varepsilon \end{cases} \quad (4)$$

where CT is a constant term that does not depend on $\mathbf{W}$ and $\mathbf{b}$; and the superscript k represents the k th iteration.

To optimize Eq. (3), an IRWLS procedure constructs a descending direction based on the previous solution and computes the next step solution along the direction. According to the Representer Theorem (Tuia et al., 2011), the best solution of minimization of Eq. (3) can be expressed as $\mathbf{w}^j = \sum_i \phi(x_i) \beta^j = \Phi^T \beta^j$, so the target of MSVR can be transformed into obtaining the best parameters of β and $\mathbf{b}$. The IRWLS procedure can be summarized by Algorithm 1 (Cantillo-Luna et al., 2022):

Algorithm 1. The IRWLS procedure.

Step I: Set $k = 0$, $\beta^k = \mathbf{0}$, $\mathbf{b}^k = \mathbf{0}$, and compute u_i^k and a_i .

Step II: Calculate the solution and label it as β^s and $\mathbf{b}^s$ based on the equation:

$$\begin{bmatrix} \mathbf{K} + \mathbf{D}_a^{-1} & \mathbf{1} \\ \mathbf{a}^T \mathbf{K} & \mathbf{1}^T \mathbf{a} \end{bmatrix} \begin{bmatrix} \beta^j \\ \mathbf{b}^j \end{bmatrix} = \begin{bmatrix} \mathbf{y}^j \\ \mathbf{a}^T \mathbf{y}^j \end{bmatrix}, \quad j = 1, \dots, Q \quad (5)$$

(continued on next column)

(continued)

where $\mathbf{a} = [a_1, \dots, a_l]^T$; $\mathbf{D}_a$ is defined as $\text{diag}(a_1, \dots, a_l)$; $\mathbf{1}$ is a column vector of l ones; and $(\mathbf{K})_{ij} = \kappa(x_i, x_j)$ is kernel matrix.

Step III: Define a descending direction:

$$\mathbf{p}^k = \begin{bmatrix} \mathbf{W}^s - \mathbf{W}^k \\ \mathbf{b}^s - \mathbf{b}^k \end{bmatrix}^T \quad (6)$$

where $\mathbf{W}^s$ and $\mathbf{b}^s$ are constructed with Eq. (5).

Step IV: Use a backtracking algorithm to calculate β^{k+1} and $\mathbf{b}^{k+1}$ based on the descending direction, and also calculate u_i^{k+1} and a_i . Set $k = k + 1$, and return to step II until convergence.

Since the relationship between rock mass parameters and the monitoring data of the tunnel is nonlinear, the RBF kernel function suitable for high-order nonlinear problems is adopted as follows (Torabi-Kaveh and Sarshari, 2019):

$$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{r^2}\right) \quad (7)$$

where r is the kernel parameter.

The convergence proof of the above algorithm can be found in Sanchez-Fernandez et al. (2004). Because u_i^k and a_i are computed by all

Table 8

The first 50 of 130 samples of three selected rock mass parameters.

No.	Density (kg · m ⁻³)	Young's modulus (MPa)	Poisson's ratio	No.	Density (kg · m ⁻³)	Young's modulus (MPa)	Poisson's ratio
1	2100.00	350.00	0.21	26	2086.67	200.00	0.26
2	1993.33	350.00	0.21	27	2086.67	440.00	0.26
3	1993.33	110.00	0.21	28	2086.67	440.00	0.21
4	1993.33	110.00	0.27	29	1993.33	140.00	0.20
5	2020.00	260.00	0.24	30	2100.00	140.00	0.20
6	1913.33	260.00	0.24	31	2100.00	380.00	0.20
7	1913.33	500.00	0.24	32	2100.00	380.00	0.25
8	1913.33	500.00	0.29	33	1913.33	320.00	0.26
9	2100.00	440.00	0.23	34	2020.00	320.00	0.26
10	1993.33	440.00	0.23	35	2020.00	80.00	0.26
11	1993.33	200.00	0.23	36	2020.00	80.00	0.21
12	1993.33	200.00	0.29	37	2086.67	440.00	0.28
13	2006.67	290.00	0.21	38	1980.00	440.00	0.28
14	1900.00	290.00	0.21	39	1980.00	200.00	0.28
15	1900.00	50.00	0.21	40	1980.00	200.00	0.23
16	1900.00	50.00	0.27	41	1913.33	230.00	0.22
17	2006.67	290.00	0.24	42	2020.00	230.00	0.22
18	1900.00	290.00	0.24	43	2020.00	470.00	0.22
19	1900.00	50.00	0.24	44	2020.00	470.00	0.27
20	1900.00	50.00	0.29	45	1980.00	80.00	0.25
21	1966.67	350.00	0.21	46	2086.67	80.00	0.25
22	2073.33	350.00	0.21	47	2086.67	320.00	0.25
23	2073.33	110.00	0.21	48	2086.67	320.00	0.30
24	2073.33	110.00	0.26	49	2073.33	260.00	0.20
25	1980.00	200.00	0.26	50	1966.67	260.00	0.20

dimensions of $\mathbf{y}$, each individual regressor contains the information of all outputs, which improves the accuracy of MSVR (Bao et al., 2014).

2.2. Differential evolution algorithm

The differential evolution (DE) algorithm is considered one of the efficient optimization algorithms for continuous search problems (AbouOmar et al., 2022). DE has the advantages of fast speed, easy realization, and robustness, and the basic idea of DE is to adapt the search during the evolutionary process. Successful hybridization of DE with other algorithms for improving the local search capabilities is given in these studies (Chi et al., 2019; Nenavath and Jatoth, 2018). The optimization process of DE is carried out with three basic operations: mutation, crossover, and selection. The DE algorithm is described in the following subsections.

2.2.1. Initialization

The initial population of N_p vectors is randomly selected based on a uniform probability distribution for all problem decision variables D . The initial population of each individual $\mathbf{X}_s = [x_{s,1}, x_{s,2}, \dots, x_{s,p}]$ is given by (Basu, 2010):

$$x_{s,p}^0 \sim U(x_p^{\min}, x_p^{\max}) \quad (8)$$

where $s = 1, \dots, N_p$; $p = 1, \dots, D$; $x_p^{\min}$ and $x_p^{\max}$ are the lower and upper bounds of the p th decision variable; and $U(x_p^{\min}, x_p^{\max})$ represents a uniform random variable ranging over $[x_p^{\min}, x_p^{\max}]$.

2.2.2. Mutation

DE generates the new population by adding the weighted difference vector between two population members to a third member. For each target vector $\mathbf{X}_s^g$ at the g th generation, the mutant vector $\mathbf{X}_s^{g/g}$ is obtained by (Storn and Price, 1997):

$$\mathbf{X}_s^{g/g} = \mathbf{X}_a^g + F(\mathbf{X}_b^g - \mathbf{X}_c^g) \quad (9)$$

where $\mathbf{X}_a^g$, $\mathbf{X}_b^g$, and $\mathbf{X}_c^g$ are randomly selected from the population at the g th generation ($a \neq b \neq c \neq s$); and the scaling factor F controls the amount of perturbation added to the parent vector (Ilonen et al., 2003).

2.2.3. Crossover

This operation generates a trial vector $\mathbf{X}_s^{g/g}$ through random selection from the target vector $\mathbf{X}_s^g$ and the mutant vector $\mathbf{X}_s^{g/g}$ as follows (Basu, 2010):

$$\mathbf{X}_s^{g/g} = \begin{cases} \mathbf{X}_s^{g/g}, & \text{if } \rho \leq P_{cr} \\ \mathbf{X}_s^g, & \text{otherwise} \end{cases} \quad (10)$$

where ρ is a uniformly distributed random number within $[0, 1]$; and P_{cr} represents a probability factor that controls the population diversity.

2.2.4. Selection

Perform selection for each target vector $\mathbf{X}_s^g$ by comparing its fitness with that of the trial vector $\mathbf{X}_s^{g/g}$. The vector that has the better fitness of these two vectors would survive for the next generation (Wan et al., 2019):

$$\mathbf{X}_s^{g+1} = \begin{cases} \mathbf{X}_s^{g/g}, & \text{if } \text{Fitness}(\mathbf{X}_s^{g/g}) \leq \text{Fitness}(\mathbf{X}_s^g) \\ \mathbf{X}_s^g, & \text{otherwise} \end{cases} \quad (11)$$

The process is repeated until the maximum number of iterations or the convergence criterion is reached.

2.3. Elementary effects method

For complex tunnel projects, numerical simulation is usually time-consuming, especially when the convergence is slow. The sensitivity analysis method can examine the uncertainty of the model response caused by the uncertainty in the inputs (Morris, 1991). Therefore, reducing the number of parameters with the sensitivity analysis is a useful and effective strategy that leads to an efficient back analysis method. The elementary effects method (EE) is a global sensitivity analysis method that examines the changes in model response by varying all parameters simultaneously (Saltelli, 2002). A brief introduction to EE is provided here.

The model input with d input factors x_i^v ($v = 1 \dots d$) is assumed to vary across u levels in the space of the input factors. The region of experimentation Ω is thus a d -dimensional u -level grid. For a given value of x_i^v , the elementary effect (EE) of the v th input factor is defined as (Campolongo et al., 2007):

Table 9

Monitoring data obtained from the simulation results of the first 50 samples at tunnel face advancements of 90 m.

No.	No. 1 monitoring section			No. 2 monitoring section		
	Vault settlement	Convergence	Surrounding rock stress	Vault settlement	Convergence	Surrounding rock stress
1	11.51	6.90	19.25	11.68	10.89	27.93
2	10.54	6.38	18.72	10.70	9.96	27.45
3	33.60	20.12	30.88	32.43	30.18	27.57
4	33.18	23.15	30.31	32.54	33.82	27.87
5	14.49	9.40	21.37	14.59	14.38	26.95
6	13.28	8.74	20.77	13.36	13.17	25.92
7	6.88	4.56	17.16	7.05	6.95	26.64
8	6.89	5.23	17.53	7.10	7.75	25.39
9	9.13	5.81	18.47	9.33	9.07	28.61
10	8.36	5.37	17.94	8.55	8.31	27.65
11	18.49	11.77	23.80	18.38	17.87	27.99
12	18.32	13.49	23.26	18.46	19.93	26.58
13	12.91	7.77	20.68	12.97	12.10	27.39
14	11.82	7.21	19.99	11.88	11.05	26.53
15	71.49	40.87	43.46	63.33	57.72	30.40
16	70.47	46.93	43.62	63.27	65.28	31.10
17	12.86	8.36	20.63	12.99	12.79	28.15
18	11.76	7.77	19.96	11.89	11.71	26.73
19	70.59	43.89	43.13	63.19	61.46	31.03
20	70.22	50.00	43.30	63.58	69.17	33.16
21	10.31	6.15	18.52	10.45	9.60	27.51
22	11.25	6.64	18.84	11.42	10.51	28.73
23	35.88	20.89	32.57	34.51	31.71	28.23
24	35.26	24.03	30.59	34.56	35.47	28.58
25	18.24	12.51	24.00	18.22	18.70	27.32
26	19.82	13.47	24.16	19.84	20.33	28.20
27	9.03	6.16	18.62	9.25	9.45	28.11
28	9.07	5.34	18.42	9.23	8.50	27.90
29	26.45	15.26	27.07	25.83	23.37	26.43
30	28.64	16.47	27.41	28.10	25.51	28.18
31	10.62	6.12	18.77	10.78	9.79	28.57
32	10.54	7.08	19.03	10.78	10.89	26.71
33	10.72	7.50	19.22	10.94	11.22	26.94
34	11.75	8.07	20.35	11.95	12.24	27.27
35	47.13	31.67	37.18	44.67	45.79	28.11
36	47.62	27.52	36.78	44.62	40.77	26.00
37	9.03	6.48	18.60	9.26	9.82	27.64
38	8.27	6.01	18.49	8.48	9.03	26.67
39	18.12	13.16	22.81	18.27	19.47	27.72
40	18.33	11.44	23.81	18.17	17.42	28.52
41	15.01	9.35	21.31	15.04	14.20	26.13
42	16.37	10.06	21.96	16.41	15.52	26.62
43	8.02	4.94	17.85	8.19	7.74	27.93
44	8.01	5.70	18.40	8.22	8.62	26.31
45	45.95	29.80	37.09	43.29	43.19	26.86
46	49.69	32.03	38.64	46.92	46.82	28.02
47	12.40	8.17	20.24	12.60	12.57	28.49
48	12.41	9.33	21.04	12.69	13.92	26.87
49	15.22	8.75	21.52	15.22	13.83	28.69
50	13.96	8.10	21.07	13.96	12.63	27.44

$$EE_v = \frac{y(x_i^1, \dots, x_i^{v-1}, x_i^v + \Delta, x_i^{v+1}, \dots, x_i^d) - y(x_i)}{\Delta} \quad (12)$$

where Δ is a value in $\{1/(u-1), \dots, 1-1/(u-1)\}$; u is the number of levels; and $y(\cdot)$ is the output function of model input x_i^v . The distribution of EE associated with the v th input factor is calculated by randomly sampling different x_i^v from Ω , and is denoted by $F_v \sim EE_v$. Two sensitivity measures are the sample mean μ and standard deviation σ for each input (Nikakhtar et al., 2020). μ estimates the overall influence of the input on the output. σ assesses the non-linear relation and/or the interactions with other factors. However, μ is prone to Type II error in some cases, i.e., failing in the identification of a factor of considerable influence on the model.

In order to provide a reliable ranking, Campolongo et al. (2007) proposed μ^* as a revised version of the measure μ . μ^* estimates the mean of the distribution of the absolute values of EE which is denoted by $G_k \sim |EE_k|$.

2.4. Evaluation metrics

Various measures of forecast accuracy have been proposed in the past (Chang et al., 2022). Root-mean-squared error (RMSE) is a commonly used scale-dependent measure based on squared errors. It is useful when comparing different methods applied to the same set of data (Khatib et al., 2012a). Mean absolute percentage error (MAPE) is the other measure to verify the prediction performance. MAPE has the advantage of being scale-independent (Khatib et al., 2012b). The A5 index evaluates a metric by showing the number of samples that fit the prediction values with a deviation of $\pm 5\%$ compared to real values. In this study, RMSE, MAPE, and A5 are used to accurately evaluate model performance. Specifically, the smaller the RMSE and MAPE, the more accurate the model. A5 index (Larger values are better) theoretically varies in the range $[0, 1]$. They are described as follows:

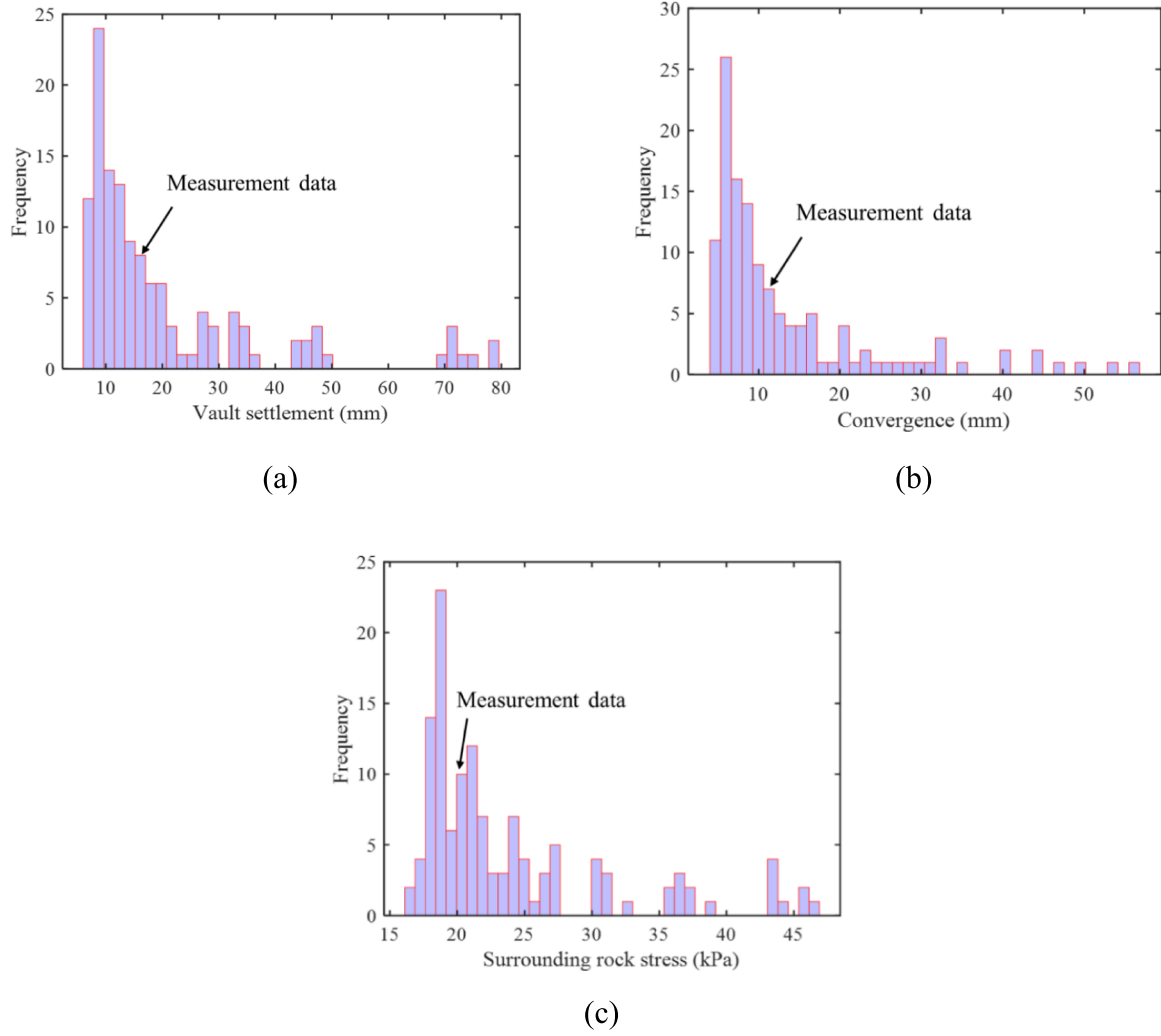


Fig. 10. Histograms of the output data from simulations for No.1 monitoring section (a) vault settlement (b) convergence and (c) surrounding rock stress.

$$\text{RMSE} = \sqrt{\frac{1}{l} \sum_{i=1}^l (y_i - \hat{y}_i)^2} \quad (13)$$

$$\text{MAPE} = \frac{100\%}{l} \sum_{i=1}^l \left| \frac{\hat{y}_i - y_i}{y_i} \right| \quad (14)$$

$$\text{A5} = \frac{1}{l} \sum_{i=1}^l \begin{cases} 1, & \text{if } \left| \frac{\hat{y}_i - y_i}{y_i} \right| \leq 0.05 \\ 0, & \text{otherwise} \end{cases} \quad (15)$$

where y_i and $\hat{y}_i$ mean the actual and estimated values, respectively; and l is the number of samples.

2.5. Framework of back analysis technique

A back analysis method based on MSVR and DE algorithm is proposed to identify rock mass parameters using monitored data during tunnel construction. The MSVR is applied to establish the mapping between rock mass parameters and the monitoring data. The DE is used to search for the optimum hyperparameter of the MSVR. The framework of the presented model consists of five steps as follows (Fig. 1):

Step I: Select an appropriate constitutive model for the rock mass and determine the lower and upper bounds of the mechanical

parameters of the constitutive model. Then, perform the numerical model using randomly generated parameters within the predefined scale, and the calculated response is stored in the dataset.

Step II: Calculate the EE of rock mass parameters and estimate the uncertainty distribution of the EE for each parameter. Rank these parameters then based on sensitivity analysis of the parameters.

Step III: Normalize the dataset between 0 and 1 and split it into a training set and a testing set. Train and test the back analysis model based on the MSVR algorithm, and use DE to search for the optimum hyperparameters of the MSVR.

Step IV: Establish the DE-MSVR model using the optimum hyperparameters of the MSVR. Input field monitoring data into DE-MSVR and calculate the mechanical parameters of the rock mass.

Step V: Perform the numerical model using the optimum mechanical parameters, and the results are compared with the field monitoring data.

3. Verification

3.1. Model setting

This section demonstrates the performance of DE-MSVR through the back-analysis of a tunnel response. The verification is performed using the simulated data from a simple circular tunnel that is modeled by FLAC 3D. The circular tunnel with a radius of 1 m that is used to test DE-MSVR is shown in Fig. 2. The tunnel is supported by a lining with a

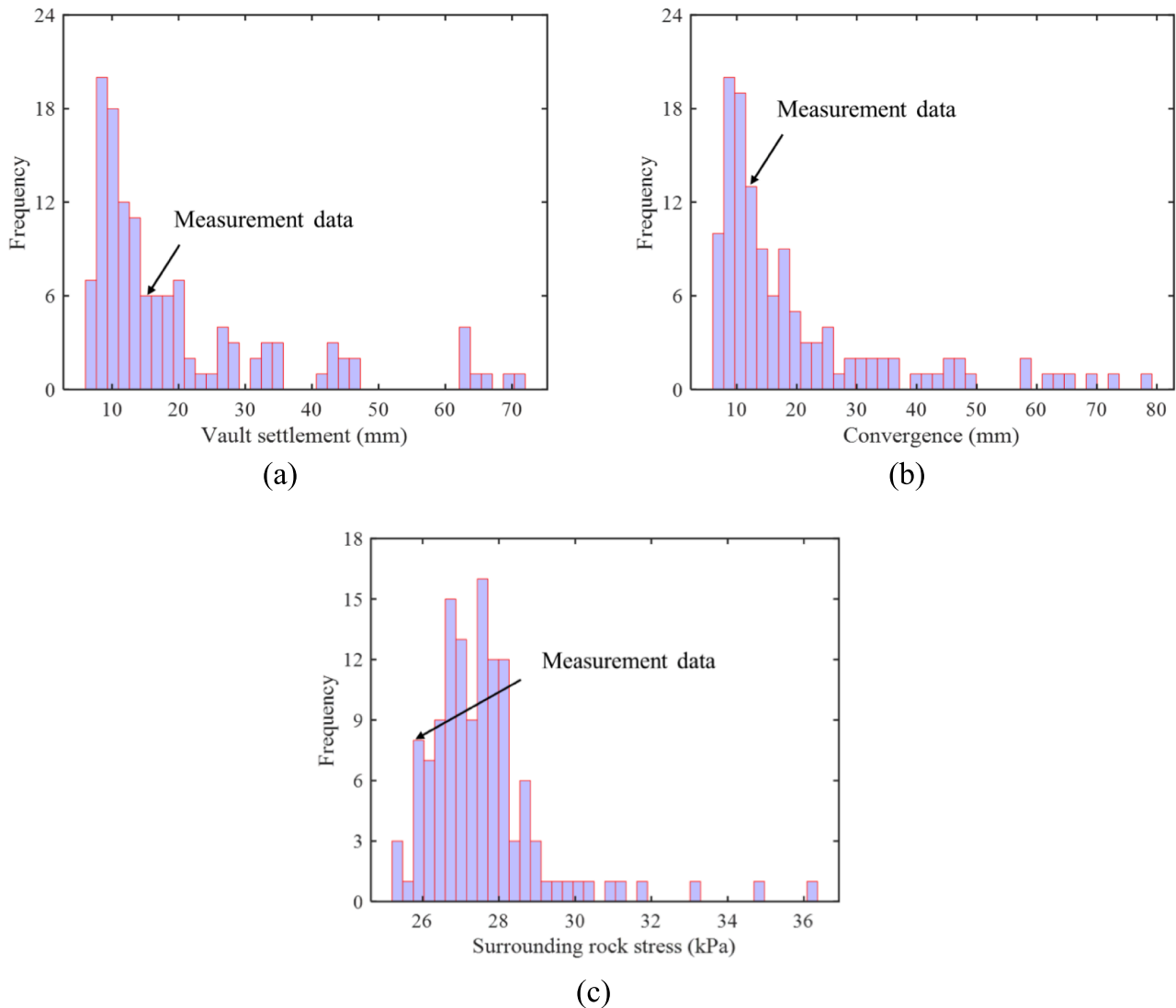


Fig. 11. Histograms of the output data from simulations for No.2 monitoring section (a) vault settlement (b) convergence and (c) surrounding rock stress.

Table 10

Training and testing results of the back analysis model for four types of monitoring data.

Data type	Surrounding rock stress	Convergence	Vault settlement	Mixed data
Training error	0.12388	0.05607	0.10221	0.01419
Testing error	0.13683	0.06071	0.10274	0.02162

Table 11

Estimation of rock mass properties for different types of monitoring data.

Data type	Rock mass parameters		
	Density (kg.m ⁻³)	Young's modulus (MPa)	Poisson's ratio
Stress of surrounding rock	2000.40	276.03	0.24870
Convergence	2004.00	231.78	0.25971
Vault settlement	2035.68	239.96	0.26148
Mixed data	2027.08	242.38	0.26665

thickness of 20 cm. To reduce the boundary effect, the whole numerical model is chosen to be $14 \times 14 \times 14$ m (Kolivand and Rahmamejad, 2018). The mesh sizes are refined near the tunnel wall and gradually increase outward. The rock mass is modeled as a homogeneous and isotropic material adhering to Mohr-Coulomb failure model. The tunnel lining is modeled as an elastic bending element that interacts with the rock mass. The properties of the tunnel lining are chosen as follows: $E = 29,000$ MPa (Young's modulus) and $\nu = 0.2$ (Poisson's ratio). With regard to the boundary conditions, the normal movements on all sides of the 3D model are restrained, whereas the bottom of the model is not allowed to move in the three directions. The model meshes and measuring points are shown in Fig. 2.

3.2. Parameter sensitivity analysis

Since the Mohr-Coulomb failure model is used to model the rock mass, five parameters that may impact tunnel response should be considered in the back analysis process, including density, Young's modulus, Poisson's ratio, cohesion, and frictional angle. To increase the computing efficiency of the back-analysis process, a global sensitivity analysis based on the EE method is employed. The parameter sensitivity powerfully depends on the range of the chosen parameter and the aim of

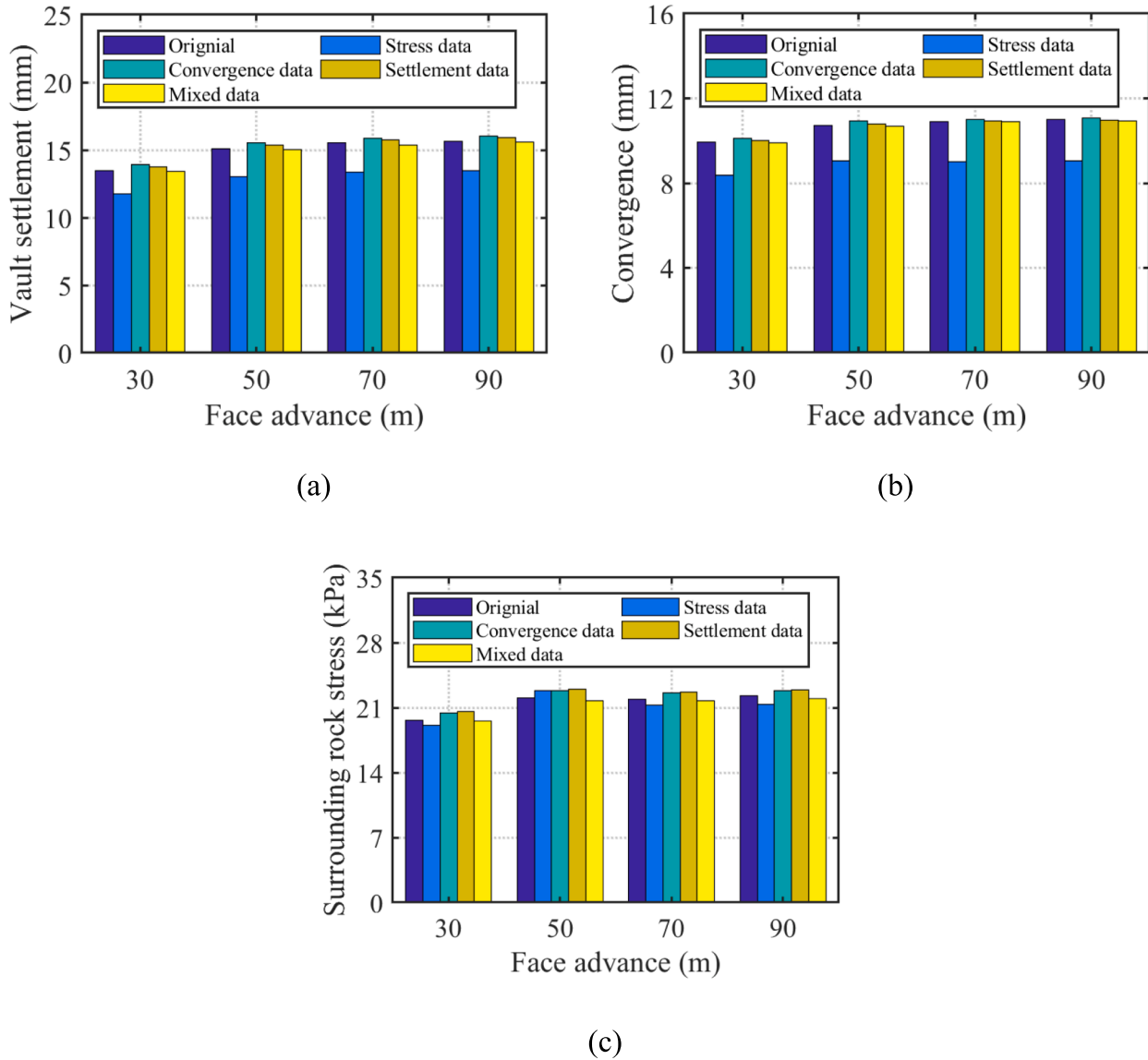


Fig. 12. Comparison between the monitoring data and predicted data from the back-analyzed parameters with different types of monitoring data for No. 1 monitoring section (a) vault settlement (b) convergence and (c) surrounding rock stress.

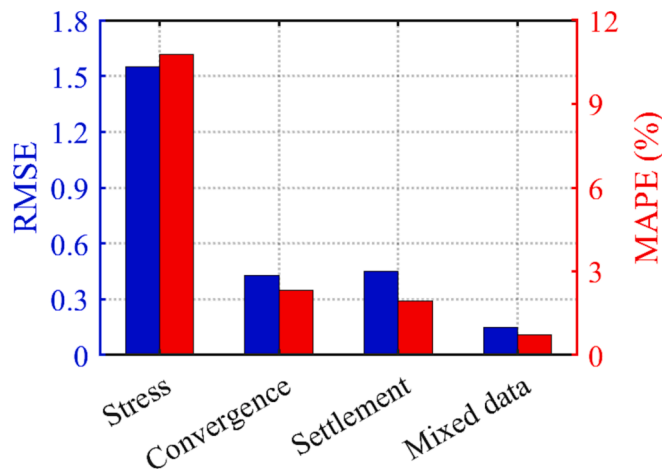


Fig. 13. RMSE and MAPE of the monitoring data and predicted data for different types of monitoring data.

Table 12

A5 of prediction results for different types of monitoring data.

Data type	Stress of surrounding rock	Convergence	Vault settlement	Mixed data
A5	70.17 %	91.67%	90.45 %	95.34%

Table 13

Training and testing results of the back analysis model for monitoring data of different advancements.

Advancement (m)	30	50	70	90
Training error	0.01647	0.01300	0.01359	0.01319
Testing error	0.02894	0.02104	0.02168	0.02162

the assessment (Ninić and Meschke, 2015). Parameter value ranges for sensitivity analysis are proposed based on the recommended values in the design phase of the Yangshan high-speed railway tunnel. These lower and upper bounds for the parameters involved in the back-analysis

Table 14
Estimation values of rock mass parameters for different advancements data.

Data type (m)	Rock mass parameters		
	Density ($\text{kg}\cdot\text{m}^{-3}$)	Young's modulus (MPa)	Poisson's ratio
30	2047.34	241.93	0.26561
50	2033.02	242.81	0.26634
70	2027.64	241.78	0.26651
90	2027.08	242.38	0.26665

are shown in Table 2.

For these given ranges of parameters, the EE method is performed to recognize the most sensitive parameters during construction. According to the method described in Section 2.3, for $u = 16$ and these five parameters, 96 numerical simulations are carried out. To evaluate the uncertainty contribution of different tunnel responses from variations of rock parameters, three kinds of monitoring data for two tunnel sections are recorded in simulations, including vault settlement, convergence, and stress of surrounding rock. The results of sensitivity analyses are shown in Figs. 3 and 4.

As shown in Fig. 3, the density, Young's modulus, and Poisson's ratio are recognized as the first three factors in the given ranges of parameters. Moreover, parameters of rock mass have different influences on three kinds of monitoring data. More specifically, for vault settlement and convergence, the density and Young's modulus are more important than the other three parameters. However, for the surrounding rock stress, besides the parameters of density and Young's modulus, Poisson's

ratio is also identified as an important parameter. Similar results can be found in the sensitivity analysis of the No. 2 monitoring section. Therefore, to increase the computing efficiency, only the density, Young's modulus, and Poisson's ratio are estimated by the back analysis technique.

3.3. Back analysis of rock mass parameters

A random parameter set is created to generate the simulated data, which is considered to represent the 'target' parameters that need to be estimated using DE-MSVR for performance verification. The values of these parameters that are used to generate the simulated data, and their lower and upper bounds are listed in Table 3. The other properties of the rock mass are chosen as follows: $F = 30^\circ$ (friction angle), $c = 90$ kPa (cohesion).

A total of 130 parameter sets for rock mass are generated to prepare the dataset for DE-MSVR (Wang et al., 2020). For each numerical model, the monitoring data is recorded for two tunnel sections. Therefore, each numerical model would produce a sample with six inputs of monitoring data (i.e., vault settlement, convergence, and surrounding rock stress) and three outputs of rock mass parameters for MSVR. The whole dataset is split into a training set (70%) and a testing set (30%). To improve the accuracy of MSVR, five nature-inspired algorithms are used to select appropriate hyperparameters, including DE, genetic algorithm (GA), particle swarm optimization (PSO), simulated annealing (SA), and artificial fish swarm algorithm (AFSA). Detailed definitions of these nature-inspired algorithms can be found in these studies (Gao, 2020;

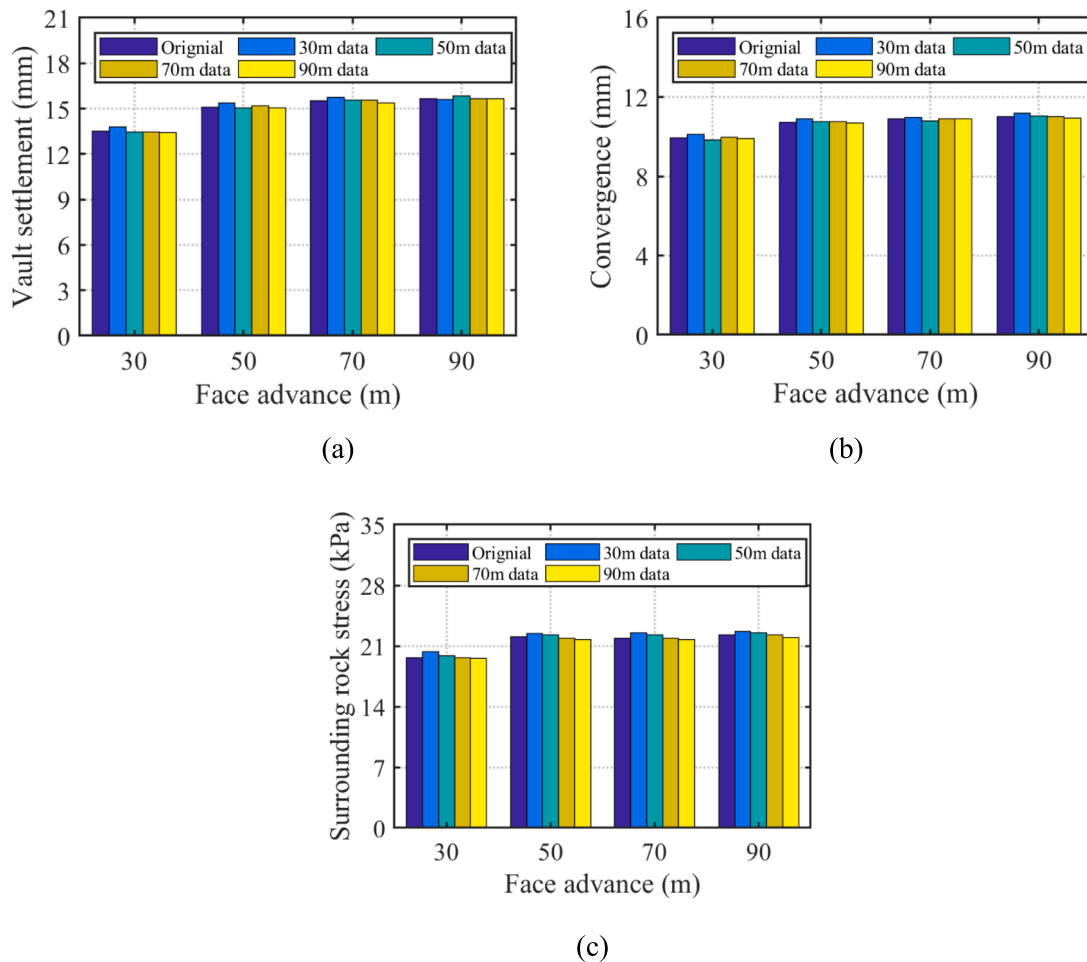


Fig. 14. Comparison between the monitoring data and predicted data from the back-analyzed parameters with monitoring data of different advancements for No.1 monitoring section (a) vault settlement (b) convergence and (c) surrounding rock stress.

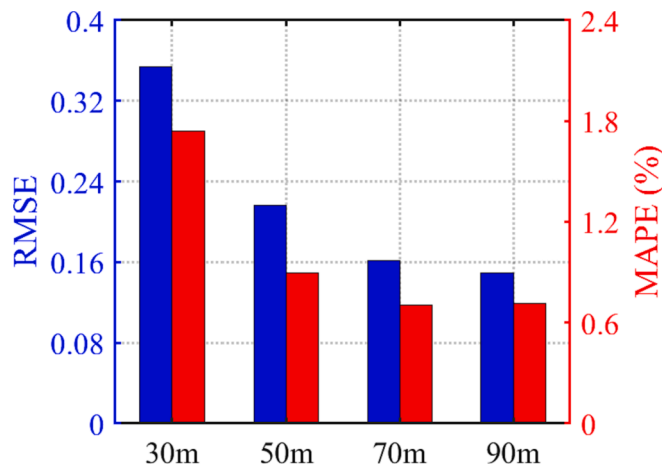


Fig. 15. RMSE and MAPE of the monitoring data and predicted data for monitoring data of different advancements.

Table 15

A5 of prediction results for different advancements data.

Data type (m)	30	50	70	90
A5	92.83%	95.73%	95.51%	95.34%

Table 16

Training results of the back analysis model for four models.

Data type	LS-SVR	GPR	DE-BPNN	DE-MSVR
Training error	0.03216	0.07361	0.03432	0.01419
Testing error	0.04368	0.08403	0.04682	0.02162

Table 17

Estimation results of rock mass properties for four models.

Method	Rock mass parameters		
	Density ($\text{kg}\cdot\text{m}^{-3}$)	Young's modulus (MPa)	Poisson's ratio
LS-SVR	2030.76	245.67	0.26736
GPR	1972.90	242.20	0.26688
DE-BPNN	2034.44	248.00	0.27188
DE-MSVR	2027.08	242.38	0.26665

Gao et al., 2016; Zhang et al., 2020). The number of individuals and the maximum number of iterations are, respectively, 100 and 1000 for all algorithms. Because of the random nature of the nature-inspired algorithm, the best result of 10 independent runs for each algorithm is selected (Gao, 2020). Table 4 shows the training and testing results of MSVR for the back analysis. For the five MSVR models, the RMSE value of AFSA-MSVR is the biggest, while the RMSE values of the other methods are smaller than 0.01. With respect to the accuracy of the back analysis model, the five nature-inspired algorithms rank as follows: DE (best), SA, GA, PSO, and AFSA (worst).

Table 5 shows a comparison between the target parameters and the back-analyzed parameters. The RMSE between the targets and back-analyzed parameters is also calculated in Table 5. It can be seen from Table 5 that the RMSE values range between 0.00430 and 0.00799. It is also found that the MAPE values of all algorithms are smaller than 1%. As a result, these five algorithms provide reasonable estimations of rock mass properties. Moreover, the DE-MSVR model shows lower RMSE and MAPE values than other algorithms for both training and testing errors, which indicates that the back analysis performance of the DE-MSVR has higher prediction accuracy. This can be explained by the fact that DE has an excellent capability for solving optimization problems that require

minimization processes with non-linear and multi-modal objective functions (Basu, 2010). The RMSE, MAPE, and A5 values between the “synthetic” monitoring data and predicted data are shown in Table 6. The DE-MSVR model demonstrates superiority in estimating the rock mass parameters as indicated by lower RMSE and MAPE values and a higher A5 value for the predicted data. Hence, due to its superior performance in predicting accuracy and estimating rock mass properties, the DE-MSVR model will be employed in the estimation of rock mass parameters for a real project in the subsequent sections.

4. Applications

4.1. Project description and monitoring data

The Yangshan high-speed railway tunnel with a length of 850 m and a maximum overburden depth of 120 m is employed to illustrate the proposed approach. The excavation of this tunnel started in 2019 using NATM. The tunnel is supported by radial rock bolts with a length of 5 m and the concrete lining with a thickness of 0.2 m. The shallow section is approximately 100 m with maximum overburden of 16 m, and it is comprised of strongly and completely weathered zones and shattered fault zones. The rock mass classification of the shallow section is V grade with broken and poor stability. The monitoring data of vault settlement, convergence, and stress of surrounding rock is important information during construction, which directly reflects the tunnel status. An automatic monitoring system for two sections of the tunnel is installed to obtain the monitoring data. To ensure construction safety and quality, monitoring data is recorded from the automatic monitoring system over 30 days. The typical cross-section and three measuring points for a tunnel section are shown in Fig. 5. Fig. 6 shows the installation of monitoring instruments during the construction of the Yangshan tunnel. Fig. 7 shows the monitoring data for two monitoring sections with tunnel face advancement.

4.2. Numerical model and parameters

A finite-difference model for the shallow section of the Yangshan tunnel is simulated by FLAC 3D. To reduce the boundary effect, the whole numerical model is chosen to be $100 \times 100 \times 40$ m (Kolivand and Rahmancejad, 2018). Based on the results of the parameter sensitivity analysis, the density, Young's modulus, and Poisson's ratio should be estimated by the back analysis technique. The lower and upper bounds of rock mass parameters (same as those in Table 1) are recommended values in designing the Yangshan high-speed railway tunnel. The average values of the other rock mass properties are chosen as follows: $F = 30^\circ$ (friction angle), $c = 90$ kPa (cohesion). Rock bolts are modeled as cable elements, and the concrete lining is modeled as shell elements. The mechanical parameters of the tunnel support are shown in Table 7. Fig. 8 presents the numerical model and measuring points for the back analysis.

A simulation case is carried out to evaluate the reliability of the finite difference model for the shallow section of the tunnel. The initial values of the rock mass parameters correspond to the average values of their parameter ranges. Note that simulated monitoring data is collected after the monitoring section has been excavated to ensure consistency with monitoring data from the actual excavation. Fig. 9 shows the comparison between the numerical solution and the measuring data at the same location. The numerical solution exhibits a similar trend compared to the measuring data, which indicates the finite difference model has good performance. However, it is easily observed that significant residuals exist, which means it is necessary to employ back analysis techniques to make the values of the rock mass parameters close to real ones.

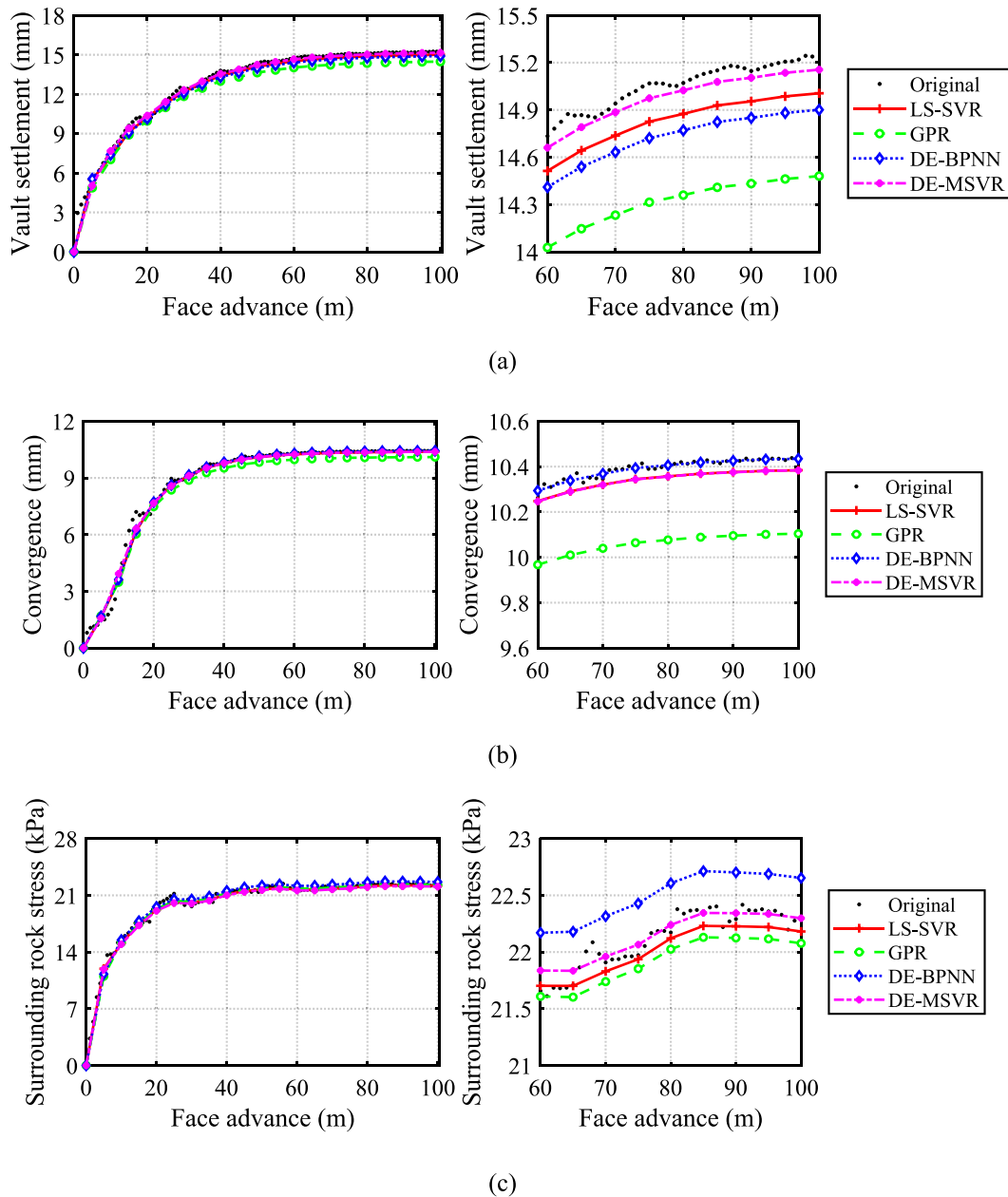


Fig. 16. Prediction results of monitoring data for No.1 monitoring section by four models (a) vault settlement (b) convergence and (c) surrounding rock stress.

4.3. Parameters back analysis

4.3.1. Numerical dataset

Similar to the procedure in the verification (Section 3), the hyper-parameters of MSVR are determined by DE. Then, the monitoring data collected during construction is input into the optimized DE-MSVR model to estimate the rock mass properties. To obtain balanced sample data, sampling for uniformity (SU) is used, which was developed based on the principles of meeting close-to-theoretical parameter distributions and maximizing trajectory spread (Khare et al., 2015). Table 8 presents the first 50 samples among the 130 available for three selected rock mass parameters. For each numerical model, the monitoring data (i. e., vault settlement, convergence, and surrounding rock stress) are then recorded to train the DE-MSVR model. Table 9 also shows the monitoring data gathered from the simulation results of the first 50 samples as tunnel face advancements of 90 m are made.

Two monitoring sections are selected to calculate the histograms of the output data from simulations in Figs. 10 and 11. The histograms

reveal that the monitoring data are not uniformly distributed, but are primarily concentrated on smaller values. The measurement data obtained during the construction of the Yangshan high-speed railway tunnel have been indicated in Figs. 10 and 11. A clear observation is that not all measurement data fall into the range of higher data frequency. Consequently, the quality of the training data is sufficient to estimate the parameters of the rock mass.

4.3.2. Parameters back analysis with different types of monitoring data

To obtain accurate rock mass properties, four types of monitoring data are used to train the DE-MSVR model, including vault settlement data, convergence data, stress data, and mixed data of deformation (vault settlement and convergence data) and stress. Table 10 shows the training and testing results of DE-MSVR for the back analysis. The RMSE value of DE-MSVR with surrounding rock stress data is larger than that of DE-MSVR with deformation data, which indicates that the deformation data shows better performance. Moreover, the DE-MSVR with mixed data has the best prediction results with the lowest RMSE value.

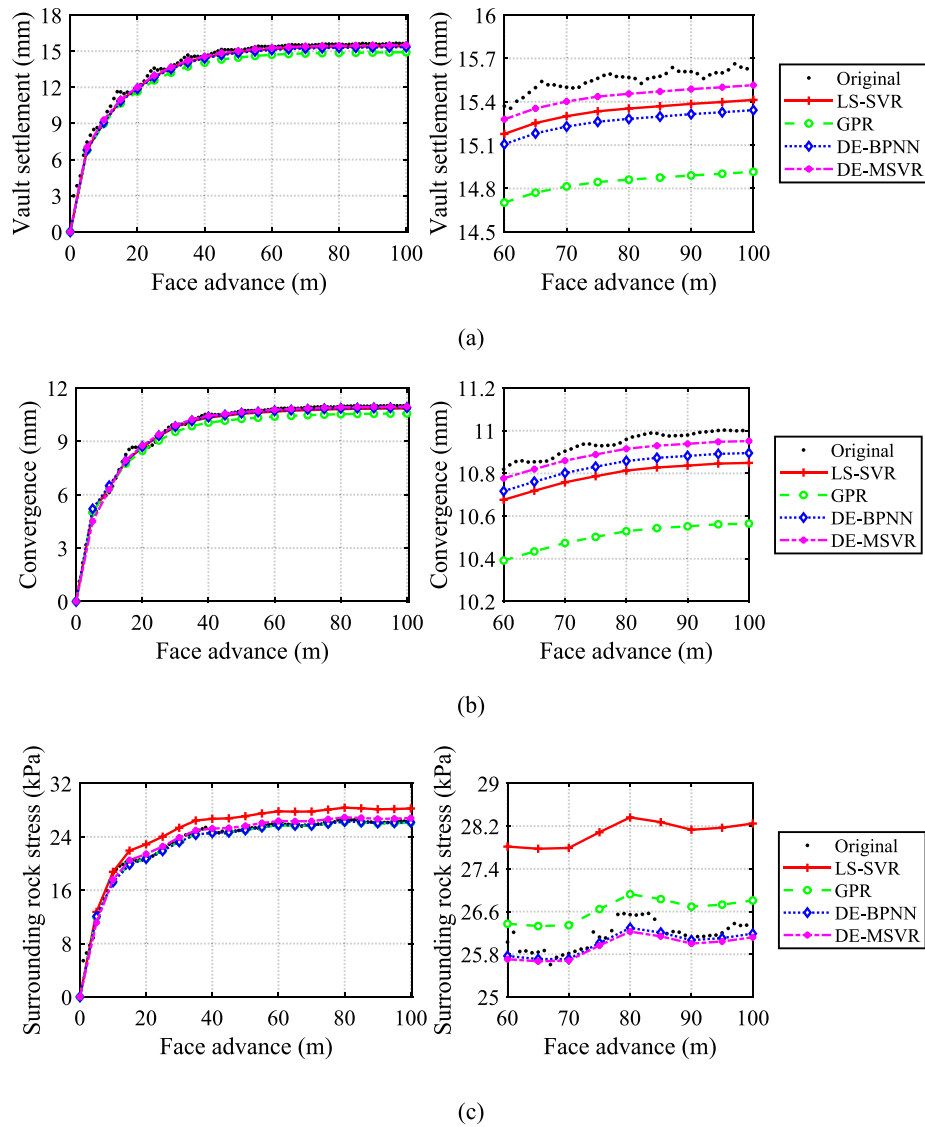


Fig. 17. Prediction results of monitoring data for No.2 monitoring section by four models (a) vault settlement (b) convergence and (c) surrounding rock stress.

Table 18

A5 of prediction monitoring data for four models.

Model	LS-SVR	GPR	DE-BPNN	DE-MSVR
A5	77.34%	85.83%	91.50%	95.34%

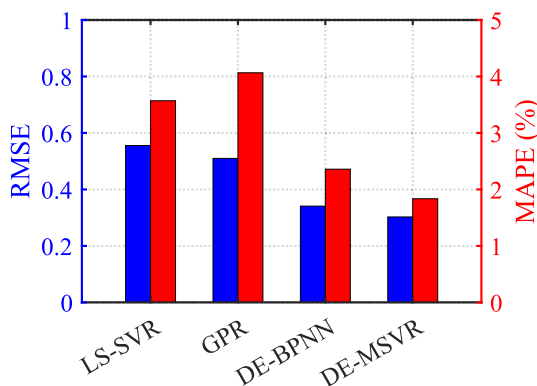


Fig. 18. Comparison results of prediction monitoring data for four models.

Thus, the mixed data can model the relationship between the monitoring data and rock mass properties with more satisfactory prediction accuracy.

Table 11 shows an estimation of rock mass properties for different types of monitoring data. For brevity, a comparison between the monitoring data and predicted data from the back-analyzed parameters from DE-MSVR for the No.1 monitoring section at different face advancements is shown in Fig. 12. The RMSE and MAPE values between the monitoring data and predicted data are shown in Fig. 13. Table 12 also shows A5 values between the monitoring data and predicted data. It is found from Fig. 12 that the predicted data exhibits a notably good agreement with the monitoring data for four models. However, the RMSE and MAPE values of DE-MSVR with mixed data are the lowest in Fig. 13. Thus, the DE-MSVR model with mixed data exhibits superiority in estimating rock mass properties.

4.3.3. Parameters back analysis with monitoring data of different advancements

To obtain accurate parameters of the rock mass at the early stages, four training samples at tunnel face advancements of 30 m, 50 m, 70 m, and 90 m are used to train the DE-MSVR model, respectively. Table 13 shows the training and testing results of DE-MSVR. The DE-MSVR model

of tunnel face advancement of 30 m has the poorest prediction results with the highest RMSE. Moreover, the RMSE values of the other three models are smaller than 0.015 and have similar results for both training and testing results. This is because the monitoring data gradually stabilize as the excavation face advances.

Table 14 shows an estimation of rock mass properties for monitoring data from different advancements. A comparison between the monitoring data and predicted data from the back-analyzed parameters from DE-MSVR for the No. 1 monitoring section is shown in Fig. 14. The RMSE and MAPE values between the monitoring data and predicted data are shown in Fig. 15. Table 15 also shows A5 values between the monitoring data and predicted data. The RMSE and MAPE values of DE-MSVR gradually decrease with tunnel face advancement, indicating that the stability of monitoring data over time improves the performance of the back analysis model. Similar results can be found in terms of the A5 index. The DE-MSVR with data for tunnel face advancement of 30 m has the maximum RMSE and MAPE values, which are 0.35 and 1.74%, respectively. In addition, it is also seen that, except for the DE-MSVR with data for tunnel face advancement of 30 m, the other three models have similar accuracies in estimating rock mass properties. Hence, it is feasible to use the monitoring data at the early stages combined with the numerical simulation for parameter back analysis.

4.4. Comparison of different parameter estimation methods

To further verify the performance of the DE-MSVR model, the following models will be used for comparative purposes: (1) a back-propagation neural network (BPNN); (2) a Gaussian process regression (GPR); and (3) a least-squares support vector regression (LS-SVR). BPNN is a forward feedback network, which is also a commonly used neural network, especially in geotechnical structures (Fei et al., 2020). DE is used to search for the optimum hyperparameter of BPNN. GPR is a nonparametric probabilistic method to solve nonlinear and complex functions in a dataset (Khatib et al., 2012b). LS-SVR replaces the convex quadratic programming problem of SVR with a convex linear system and largely decreases the training time (Zhu and Gao, 2018). This method is suitable for multi-output regression problems, especially when the outputs correlate with each other (Xu et al., 2014). All of these methods use the same learning settings as those proposed in this paper. The training results and estimation results of rock mass properties for these methods are provided in Tables 16 and 17. GPR has the poorest accuracy with the highest RMSE. This proves that GPR usually requires more training samples to improve performance for complex nonlinear problems (Lu et al., 2021). Moreover, DE-BPNN and LS-SVR have similar training and testing results. Compared with GPR and DE-BPNN, DE-MSVR has the lowest RMSE value. It can be concluded that the DE-MSVR model exhibits superiority over the other three methods.

A comparison between the monitoring data and predicted data from the back-analyzed parameters from these methods is shown in Figs. 16 and 17. To quantify the performance of these methods further, RMSE and MAPE are calculated and compared in Fig. 18, and A5 index is also calculated and shown in Table 18. As shown in Figs. 16, 17, and 18, GPR has the poorest prediction results with the highest RMSE and MAPE values. It is also seen that the GA-BPNN and LS-SVR show lower RMSE and MAPE values than GPR, which indicates that these two models have better prediction accuracies. However, Table 18 shows that the LS-SVR has the lowest A5 value, which indicates poor prediction results. Moreover, it is easily observed that DE-MSVR has better performance than the other three models. From all of the above analyses, the DE-MSVR could be a promising technique for estimating rock mass properties.

5. Conclusion

In this paper, a new back analysis method based on multi-output support vector regression (MSVR) and differential evolution (DE)

algorithms has been developed to estimate rock mass properties. Monitoring data from the automatic monitoring system of a high-speed railway tunnel for one month is employed to demonstrate the feasibility of the presented approach. The following conclusions are obtained: (1) The DE-MSVR model provides higher prediction accuracy than other nature-inspired metaheuristic optimization algorithms in estimating rock mass properties based on monitoring data. (2) Mixed data of the deformation (i.e., vault settlement and convergence data) and stress show better performance in modeling the relationship between monitoring data and rock mass properties. (3) It is feasible to use monitoring data at the early stages of tunnel excavation combined with numerical simulation for back analysis of rock mass parameters. (4) DE-MSVR outperforms the backpropagation neural network, Gaussian process regression, and least-squares support vector regression models in predicting rock mass properties due to its superior generalization capacities. However, the study suffers some limitations that will be addressed in the future, such as the limited amount of monitoring data used to train the models and the lack of consideration for dynamic factors that may affect the behavior of rock mass properties.

CRediT authorship contribution statement

Xiangyu Chang: Conceptualization, Methodology, Software, Validation, Writing – original draft. **Hao Wang:** Supervision, Funding acquisition, Investigation, Writing – review & editing. **Yiming Zhang:** Methodology, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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